

ADITYA DNYANESHWAR DAHALE

Systems & Infrastructure Software Engineer | Rust, C++, Go, Linux Internals, Distributed Systems

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VERIFIED MERGED OPEN-SOURCE CONTRIBUTIONS (ACCEPTED & SHIPPED)

- **Tokio Async Runtime (tokio-rs/tokio)** | [PR #8405](#) — *Semaphore Integer Overflow & Permit Leak* ✓ MERGED
Resolved critical permit leak in SemaphorePermit::merge where counts exceeding u32::MAX wrapped and leaked permits on drop; converted storage to safe checked usize.
- **FoxIO JA4 Fingerprinting (FoxIO-LLC/ja4)** | [PR #312](#) & [#314](#) — *Zero-Allocation Header Filter & Raw Hex Digest* ✓ 2x MERGED
Eliminated heap string allocations in HTTP/1 header filtering (#312); optimized hash12 by hex-encoding 6-byte raw digest slices, cutting 2 allocations and avoiding encoding 26 unused bytes (#314).
- **smoltcp TCP/IP Wire Stack (smoltcp-rs/smoltcp)** | [PR #1194](#) — *Zero-Allocation Wire Option Parser* ✓ MERGED
Engineered single-pass zero-allocation TcpOptionsSummary wire parser for standalone embedded network stack, removing heap allocations on the packet wire-path to boost packet ingestion.
- **SmartCore ML Engine (smartcorelib/smartcore)** | [PR #386](#) — *Pre-Allocated Capacity in Ensemble Trees* ✓ MERGED
Optimized ensemble tree training cycles by pre-allocating exact vector capacities in Random Forest models, eliminating repeated heap memory reallocations.
- **MaxMindDB Reader (oschwald/maxminddb-rust)** | [PR #120](#) — *Native Big-Endian Byte Decoding* ✓ MERGED
Accelerated GeoIP binary lookups using native big-endian byte decoding (u32::from_be_bytes), enabling single-CPU-instruction decoding of 32-bit database nodes.

PARADIGM CORE ETHEREUM INFRASTRUCTURE (RETH • ALLOY • FOUNDRY)

- **Paradigm Reth (paradigmxyz/reth)** | [PR #27049](#) — *Snapshot Manifest Validation & Zero-Division Bounds* Open • Review Requested
Hardened SnapshotManifest::validate() to reject malformed manifests with blocks_per_file == 0 prior to expansion. Guarded ChunkedArchive::num_chunks() with saturating math to prevent snapshot panics during node sync.
- **Alloy Core SDK (alloy-rs/alloy)** | [PR #4191](#) — *Pending Transaction Heartbeat Recovery* Open • Review Requested
Fixed silent stream freeze in PendingTransactionBuilder::watch when pollers skip mined blocks during network jitter. Scans skipped block ranges across heartbeats to guarantee pending transactions complete.
- **Foundry (foundry-rs/foundry)** | [PR #16707](#) — *Chisel Session Isolation & Path Traversal Guard* Open • Review Requested
Hardened session validation in crates/chisel against Windows NTFS Alternate Data Streams (chisel-foo:bar.json) and path traversals, preventing arbitrary disk write escapes.

TIER-1 SYSTEMS RUNTIME & CLUSTER INFRASTRUCTURE (HYPER • FLY.IO • ASTRAL • PINGORA)

- **Hyper HTTP Engine (hyperium/hyper)** | [PR #4183](#) — *RFC 6585 max_header_size Protocol Enforcement* Open • In Review
Added server/client builder APIs, strict RFC 6585 \$5 (431 Too Large) rejection, stream buffer zero-copy guards, and chunked trailer decoder limits across 180+ tests.
- **Fly.io Cluster Infra (superfly/corrosion, ltx, macaroon)** | [PRs #578, #102, #104, #48](#) — *Distributed Engine Resilience* In Review / CI Passed
Eliminated schema migration crash panics in Corrosion (#578); fixed zero-page checksums in LTX (#104); enforced decoder bounds (#102); eliminated flaky deadline backoff bounds in Macaroon auth (#48).
- **Astral uv Toolchain (astral-sh/uv)** | [PR #21457](#) — *Cross-Process File Lock Timeout Propagation* CI 100% Green
Decoupled lock acquisition from static timeouts by threading configurable options through EnvironmentOptions; verified across 55 CI matrix suites across 11 OS/arch targets.
- **Cloudflare Pingora (cloudflare/pingora)** | [PR #996](#) — *RFC 9110 Upstream Timeout Mapping* CI 100% Green
Resolved upstream timeout classification in Cloudflare's core reverse proxy; mapped connect, TLS handshake, and read/write timeouts to 504 Gateway Timeout instead of generic 502 Bad Gateway.
- **Turso libSQL (tursodatabase/libsql)** | [PR #2280](#) — *Default WAL Journal Concurrency* CI 100% Green
Enforced Write-Ahead Logging (WAL) concurrency as default for local SQLite instances, preventing database locks under concurrent access. All 12 CI suites passed.
- **Google Comprehensive Rust (google/comprehensive-rust)** | [PR #3275](#) — *LLVM Undefined Behavior Modeling* CI Green • CLA Signed
Accurately modeled language-level Undefined Behavior using intrinsic i32::unchecked_add in unsafe courseware modules per Google Staff Engineer design review.

FEATURED SYSTEMS PROJECTS & ARCHITECTURES

- **Spryzn+ — Zero-Allocation Network Security & Wire Inspection Engine** | *Rust, Lock-Free Ring Buffers, SIMD, Tokio, Rayon Bulkheads, eBPF/XDP, Docker*
 - **4.83M RPS Throughput & 207 ns P99 Latency:** Zero dynamic heap allocations on wire path via borrowed stack slices (Cow::Borrowed) under high-throughput packet ingestion.
 - **27.99 ns Vectorized Normalization:** 256-byte compile-time LUT (NEEDS_DEEP_NORM_TABLE) with single-instruction vectorized byte lookups, bypassing expensive regex engines.
 - **Lock-Free Concurrency & Rayon Bulkheads:** Atomic ring queues eliminating thread mutexes; isolated jumbo payloads (>8KB) to Rayon workers to guarantee Tokio event loop availability.
 - **Benchmarking & Profiling:** Packaged with automated Criterion benchmarking suite, Docker multi-stage harness, and memory profiling for L1/L2 cache locality.

AGENTIC OPERATIONS & AI-NATIVE SYSTEMS INFRASTRUCTURE

- **AI-Native Engineering & Agent Tooling:** Built custom MCP (Model Context Protocol) servers and deterministic agent orchestration tools enabling coding agents to safely investigate test flakes, execute build verification, and automate release pipelines.
- **Observability & Safe Guardrails:** Deployed Prometheus, Grafana, and Loki telemetry stacks with automated alerting guardrails; codified operational runbooks into programmatic verification harnesses to minimize blast radius.

TECHNICAL TOOLBELT & SYSTEMS PROFICIENCIES (ATS-OPTIMIZED)

- **Languages:** Rust (Advanced / Unsafe / Systems), C/C++ (C++17/20), Go, Python, Bash, SQL, x86_64 Assembly fundamentals
- **Systems & Architecture:** Zero-Allocation Wire Parsing, Lock-Free Ring Buffers, SIMD / LUTs, Rayon Bulkheads, eBPF/XDP, Tokio Runtime, Linux Epoll, Memory Profiling (perf, Valgrind, Criterion)
- **Infrastructure & DevOps:** Linux Kernel & systemd, Docker, Kubernetes, CI/CD Workflows, Prometheus, Grafana, Loki, Terraform, MCP Servers
- **Protocols & Distributed:** TCP/IP Wire Stacks (smoltcp), HTTP/1 & HTTP/2 (RFC 9110/6585), TLS 1.3 / JA4 Fingerprinting, BFT Consensus, Reverse Proxies

EDUCATION

Savitribai Phule Pune University (SPPU) — Bachelor of Engineering (B.E.) in Computer Engineering

Pune, India

Cumulative SGPA: 8.32 / 10.0 | Core Focus: Systems Programming, Operating Systems, Computer Architecture, Data Structures